



SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2022-2023

Program/s: B.Tech/MBA Tech

Year: II Semester: IV

Stream/s ;Data science

Subject: Introduction to data, signal and image analysis

Time: 03 hrs (10:00 am to 1:00 pm)

Date: 01 / July / 2023

No. of Pages: 03

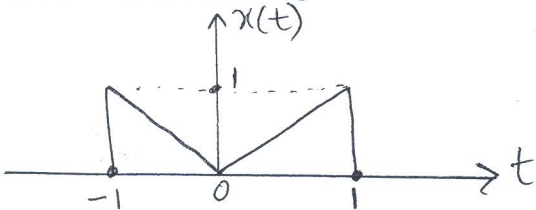
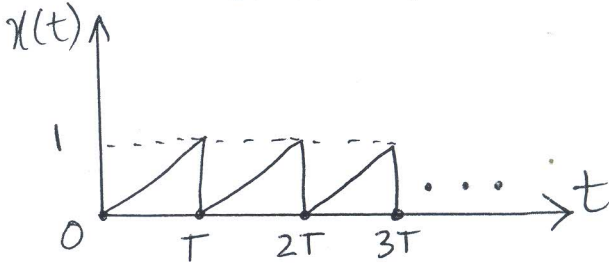
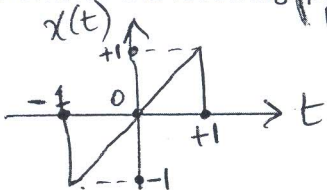
Marks: 100

Re- Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	marks
CO1- ; BL-3	a.	If $X(S) = \frac{2s+6}{s(4s+2)}$ then find $x(0)$ and $x(\infty)$	5
CO-4 ; BL-6	b.	5X5 image is defined by the following expression: $f(x,y) = x^2 - y^2 $ Construct the image, determine the size of the image. Develop a new image by only retaining row and column number 0,2 and 4	5
CO4- ; BL-1	c.	What is bit plane slicing? Find the bit planes of the given 3X3 image $A = \begin{bmatrix} 2 & 6 & 6 \\ 6 & 7 & 7 \\ 1 & 2 & 4 \end{bmatrix}$	5
CO-3 ; BL-2	d.	$x(t) = 2 \cos 5\pi t + 7 \cos 15\pi t + 5 \cos 10\pi t$. Determine min sampling rate and corresponding discrete signal	5

Q2 CO1-; BL-3	a	Find Fourier Transform for the following: - 	10
Q2 CO1-; BL-5	b	Find the Z transform of u(n) and r(n). Further, evaluate Z [sin wn. U(n)]	10
Q3 CO-1; BL-3	a	Determine Laplace Transform for the following (non periodic) waveform: 	10
Q3 CO1-; BL-2	b	What are analog and digital signals? Write a detailed note on them.	10
Q4 CO1-; BL-3	a	Evaluate Laplace Transform for the following (periodic) waveform  NOT periodic	10
Q4 CO1-; BL-1	b	Describe the simplified model of a data communication system. Present all possible operations (operation types) which can be performed on a cts with examples	10
Q5 CO-3; BL-2	a	What is convolution? Calculate y(n) if x(n)= {1,2,3} and h(n) = {2,1} using circular convolution	10
Q5 CO-1; BL-4	b	Perform $L\left[\frac{e^{-2t} \cdot \sin 2t \cdot \cos ht}{t}\right]$. Also discuss applications of Laplace Transform.	10
Q6 CO1-; BL-5	a	What are the properties of Roc of Z transform? If $X(Z) = \frac{Z^2}{(Z^2 - 3Z + 2)}$ Find and Plot x(n) for all possible ROCs	10

Q6 CO2-; BL-4	b	Distinguish between FIR and IIR filters	10																									
Q7 CO4-; BL-6	a	What is histogram equalization? Perform histogram equalization on the following 5 x 5 image. Plot histogram of original image and equalized image. Comment on the results obtained after equalization. <table><tr><td>6</td><td>6</td><td>7</td><td>7</td><td>6</td></tr><tr><td>5</td><td>2</td><td>2</td><td>3</td><td>4</td></tr><tr><td>3</td><td>3</td><td>4</td><td>4</td><td>5</td></tr><tr><td>5</td><td>7</td><td>3</td><td>6</td><td>2</td></tr><tr><td>7</td><td>6</td><td>5</td><td>5</td><td>4</td></tr></table>	6	6	7	7	6	5	2	2	3	4	3	3	4	4	5	5	7	3	6	2	7	6	5	5	4	10
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Q7 CO1-; BL-5	b	Find $L[e^{-at} \cdot \cos wt \cdot u(t)]$. Also find Laplace of a standard ramp signal $r(t)$	10																									